

CSE 280 Challenge Set 04

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Question 1

For each of these functions, determine if they are well-defined, one-to-one, onto and/or a bijection. For these problems, let $A = \{x \mid x \in \mathbf{Z}, 0 < x < 5\}$ and let $B = \{x \mid x \in \mathbf{Z}, 0 \leq x \leq 5\}$. You may want to draw a digraph of each function (with dots and arrows). Recall that if not well-defined, then it is neither 1-1, onto, nor a bijection.

Function	Well Defined	1-1	Onto	Bijection
$f : A \rightarrow A$, where $f = \{(1, 2), (2, 2), (3, 3), (4, 3)\}$				
$f : A \rightarrow A$, where $f = \{(1, 1), (2, 2), (3, 4), (4, 3)\}$				
$f : A \rightarrow A$, where $f = \{(2, 2), (3, 4), (4, 3)\}$				
$f : A \rightarrow B$, where $f = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$				
$f : B \rightarrow A$, where $f = \{(0, 1), (1, 1), (2, 2), (3, 2), (4, 4), (5, 3)\}$				

Question 2

For each of these functions, determine if they are well-defined, one-to-one, onto, and/or a bijection. Consider using a graphing calculator like <https://www.desmos.com>.

Function	Well Defined	1-1	Onto	Bijection
$f : \mathbf{Z} \rightarrow \mathbf{Z}$, where $f(x) = 5x - 7$				
$f : \mathbf{Z}^+ \rightarrow \mathbf{Z}^+$, where $f(x) = \lceil \frac{x}{5} \rceil$				
$f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = x^5 - 3x$				
$f : \mathbf{R}^+ \rightarrow \mathbf{R}$, where $f(x) = \frac{1}{x}$				
$f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = \log_5(x^2)$				
$f : \mathbf{R}^+ \rightarrow \mathbf{R}$, where $f(x) = \log_5(x^2)$				
$f : \{x \mid x \in \mathbf{R}, x \neq 0\} \rightarrow \mathbf{R}$, where $f(x) = \log_5(x^2)$				

Question 3

Part 1

Find the inverse $f^{-1}(x)$ (if it exists) of the following functions:

- $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = (x - 32) * 5/9$
- $f : \mathbf{R} \rightarrow \mathbf{R}^+$ where $f(x) = \frac{1}{2^x}$

- $f : \mathbf{R} \rightarrow \mathbf{Z}$, where $f(x) = \lfloor x \rfloor$
- $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = x^2$
- $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = x^3$

Part 2

Select a value of x (per the domain) and calculate $(f^{-1} \circ f)(x)$ for all functions that had an inverse in Part 1 to demonstrate that the inverse was successfully created.

Question 4

Define the following two functions:

- $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = \lceil x \rceil - x$
- $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = x - \lfloor x \rfloor$

Part 1

Graph both functions f and g using a graphing calculator. Explain what this so-called "sawtooth" function represents.

Part 2

Based on your answer above, does an inverse to f exist?

Part 3

Determine what $f(x) + g(x)$ is equal to.